pySeqRNA: An automated Python package for RNA sequencing data analysis

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Objectives

* Python-based end to end RNAseq analysis pipeline
* Reference-based analysis
* *De novo* assembly-based analysis
* Slurm job scheduling on HPC
* Python subprocess integration
* Different read mapping-based count approach
* User end customizable

Input

A user customized input config file with Sample Name, Replication, Identifier and comments

List of Comparison between Samples for Differential expression analysis

```
[input_info]
;Sample_Name  Replication  Identifier   File_Path
Mock.1h.A   =    M1A    M1      ./data/SRR446027_1.fastq.gz
Mock.1h.B   =    M1B    M1      ./data/SRR446028_1.fastq.gz
Avr.1h.A    =    A1A    A1      ./data/SRR446029_1.fastq.gz
Avr.1h.B    =    A1B    A1      ./data/SRR446030_1.fastq.gz
Vir.1h.A    =    V1A    V1      ./data/SRR446031_1.fastq.gz
Vir.1h.B    =    V1B    V1      ./data/SRR446032_1.fastq.gz
Mock.6h.A   =    M6A    M6      ./data/SRR446033_1.fastq.gz
Mock.6h.B   =    M6B    M6      ./data/SRR446034_1.fastq.gz
Avr.6h.A    =    A6A    A6      ./data/SRR446035_1.fastq.gz
Vir.6h.A    =    V6A    V6      ./data/SRR446037_1.fastq.gz
Vir.6h.B    =    V6B    V6      ./data/SRR446038_1.fastq.gz

[Combinations_for_DEGs]
Compare_samples = M1-A1, M1-V1, A1-V1, M6-A6, M6-V6, A6-V6
```

Scan me

Request pySeqRNA code

Configuration

Path to input config file
Quality trimming tool
Adapter File
Mapping tool
Read count tool
Sample Correlation Method
DEGs method type
Report type
Slurm Job scheduling
Reference Genome & GFF

```
[system_setup]
slurm=True  # True/False based on slurm
Standalone= NA  # Use only when slurm n
[Reference]
Refernce_genome_path = genome.fa # Path
Reference_GFF_path = genome.gff3 # Path
denovo_assembly = False # True only whe
[Execute_Parameter]
Path_to_input_config = input_config.ini
Quality_trimming_tool = flexbar # Check
Adapter_file = NA # Path to adapter fil
Aligner = STAR # Check documentation fo
Read_count_tool = featureCount # Check
Sample_co-relation_method = RPKM # Chec
Differential_expression_method = DESeq2
Report_type= PDF    # PDF/html default
```

QC and Trimming

FLEXBAR — flexible barcode and adapter removal

Trimmomatic

Trim Galore

De novo Assembly

Executed if reference genome not given in run_config.ini

Trinity (Inchworm, Chrysalis, Butterfly)

Quast

Corset

Read Mapping & Count

STAR
gsnap
HISAT2
Tophat2
Bowtie2

[Uniquely mapped read count]
[Multi-mapped groups]
[Multi-mapped to same Gene]

featureCounts
HTSeq
mmquant

Count & Sample Correlation

TPM
RPKM
FPKM

Normalize Counts

log_2 transformation

Structure of pySeqRNA

Config file for tools
Aligner Class
Differential expression Class
Quality and Trimming Class
Feature count class
Functional annotation Class
Class for *denovo* assembly
Sample Correlation class
Report Templates

Report

Report can be saved in PDF, HTML format

Publication ready figures and graphs

Detailed options used during analysis for each tool

Summary of results

Visualization

Quality Insights
BAM file Viewer
Read Count
Correlation Tree
HeatMap
Venn Daigram
DEGs Count
GO Enrichment
Pathway Enrichment

Functional Annotation

GOMAP
KEGG (Kyoto Encyclopedia of Genes and Genomes)

KOG Classification

biomart

SQLite

Custom Python Scripts

Differential Gene Expression

DESeq2
edgeR
DEXSeq

Python Wrapper for R packages

Comparison based analysis

Scripts to filter DEGs based on FDR and FOLD change

Contact US

SCAN ME

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Future Work

Alpha version public release
More aligner addition
Functional Annotations
More Report types
Speed up analysis
Dual-Seq RNA support
Co-expression analysis
Beta version public release
Bug Removal
Complete documentation
Jupyter notebook tutorial
Final Release